DATS 598: Data Science Capstone

Journal Entries and Progress Documentation

Statistical Truth Detection System

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Thursday 2nd October, 2025

Contents

Journal Entry - Week 3 - Statistical Truth Detection System

Date: Thursday 2^{nd} October, 2025
Duration: 10 hours

Objective

Synthesyze the literature review and understand a path forward for the truth detection project. Also summarize current project standing.

Activities Completed

- Project architecture setup (web app, tech stack, etc.).
- Literature review.
- Initial deep dive into the NLP processes needed, finding open source libraries where applicable, and determine the neccesary, nice to haves for project deliverables.

Key Findings

The literature review consisted of reading papers, open source documentation, and interacting with existing tools. The concept of fact-checking is not novel. It has been around since 2018, which coincides with the popularity rise of "fake news" and misinformation on social media platforms. As such, many fact-checking methods relate to political contexts, Wikipedia, and other trusted data sources. The following papers were most informative:

1. Remaining Product Life
2. Likelihood of Catostrophic Failure
3. Predictions of Unmeasured Data

Next Steps

- Pull in data sources for fact-checking (e.g., FEVER, HuggingFace, Wikidata, etc.).
- Begin building out the backend applications via notebook (sentence parsing and analysis, API hooks, open source and python compability and configurations for environment stability).
- Begin prototyping the web app and setting up the tech stack (confirmation of CI/CD pipelines, containerized code, frontend-backend communication).

The current tech stack (as planned, and currently built is this):

- Backend: Python, FastAPI, Uvicorn

- Frontend: JavaScript, React, Vite
- VCS/Deployment: Git, GitHub Actions for CI/CD, AWS for hosting (?)
- Database: PostgreSQL, if RAG (?)
- NLP: SpaCy, NLTK, HuggingFace Transformers
- Containerization: Docker (Docker compose)
- EDA Packages: Jupyter Notebooks, Pandas, Matplotlib, Seaborn, NLTK, SpaCy, Transformers, Torch